

CS 305: Artificial Intelligence

Course Code	CS 305
Course Name	Artificial Intelligence
L-T-P-C	3-0-0-3
Prerequisites	CS 212 Design of Algorithms or an equivalent course
Intended for	3rd and 4th year B.Tech. students
Mutual Exclusion	None
Approval	3rd Ad-hoc Senate (Nov 2012)
Revised	54th BoA (May 2024) - syllabus revised

Preamble

The Artificial Intelligence (AI) course is designed to introduce students to the foundational concepts and techniques that form the basis of intelligent computer systems. This course aims to build a strong foundational understanding of AI by covering a range of topics from heuristic search and game playing to knowledge representation and robotics. It is essential for students interested in advanced fields like machine learning, cognitive modelling, natural language processing, and computer vision.

Course Modules

- **Unit 1: Introduction to AI:** Overview, historical perspective, Turing test, agents. (3 Hours)
- **Unit 2: Search Methods and Optimization:** Problem representation, State Space Search, A* Algorithm, AO* search, Minimax and alpha-beta pruning, AI in games. (8 Hours)
- **Unit 3: Logic and Automated Reasoning:** Propositional Logic, Predicate Calculus, Resolution Refutation, Formal Systems, Soundness, Consistency, Completeness. (7 Hours)
- **Unit 4: Planning and Constraint Satisfaction:** Blocks World, STRIPS, Constraint Satisfaction Problems, Basics of Probabilistic Planning, Waltz Algorithm. (6 Hours)
- **Unit 5: Knowledge Representation and Reasoning:** Semantic Net, Frames, Scripts, Conceptual Dependency, Ontologies, Basics of Semantic Web, Intelligent Question Answering. (7 Hours)
- **Unit 6: Learning and Uncertainty:** Learning from Examples, Decision Trees, Neural Nets, Bayesian Theory, Fuzzy Logic, Non-monotonic Logic, Default Reasoning, Case-based Reasoning. (6 Hours)
- **Unit 7: Advanced Topics in AI:** Introduction to Computer Vision, Expert Systems, Natural Language Processing, Robotics, Hidden Markov Models, Reinforcement Learning. (5 Hours)

Laboratory Modules

None.

Text Books

1. Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach*. 4th Edition, Pearson Education, USA.
2. Poole, D., & Mackworth, A. (2017). *Artificial Intelligence: Foundations of Computational Agents*. 2nd Edition, Cambridge University Press, UK.

References

1. Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press, USA.
2. Luger, G. F. (2008). *Artificial Intelligence: Structures and Strategies for Complex Problem Solving*. 6th Edition, Pearson Education, USA.